Addendum: Two Adversarial Tests of the $\chi(3)$ Channel

Frobenius traces and a level-6 automorphic form against the 6N wing

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The $\chi(3)$-channel principle asserts that the 6N wing of a prime p (its class $p \bmod 3$) can enter an L-function-derived invariant only through the $q = 3$ Euler factor $\chi_p(3)$. A principle of this kind earns its standing by surviving attempts to break it. This addendum reports two such attempts—deliberately constructed to give the wing its best chance—together with the data that defeat them. They fail by *opposite* mechanisms and reach the *same* verdict.

The two break-in attempts

Test A — decrepitation (Galois fluid). Bombard the curve $E_p : y^2 = x^3 - p$ at a prime "temperature" ℓ and extract the Frobenius trace $a_\ell(E_p)$, for $\ell = 7, 13$, over all 2260 primes $3 < p < 2 \times 10^4$. The hope: a quantity drawn from deep in the Galois representation might carry wing information past the CRT lattice.

Test B — a native level-6 plume (Langlands). Instead of a generic object, choose one whose *level* already carries 6: the cusp-form newforms of $\Gamma_0(6)$, weights $k = 4, 6$ (the space $S_2(\Gamma_0(6))$ is 0-dimensional, so $k = 2$ is empty). Extract the Hecke eigenvalue a_p at all 667 primes $3 < p < 5000$. The hope: an automorphic form built on 6 might split its spectrum along the $6N$ wings.

The data's verdict

Both attempts return a flat null on the wing, with effect sizes below the noise floor:

	Test A: $a_\ell(E_p)$	Test B: a_p of level-6 newform
Wing effect (Cohen's d)	-0.006 (a_7), -0.004 (a_{13})	$+0.028$ (k=4), $+0.043$ (k=6)
Shuffle test	null	$p = 0.625,\ 0.430$
Controller of the observable	$p \bmod \ell$	none of small modulus
Variance explained by it	$\eta^2 = 1.000$	$\eta^2(p \bmod 12) \approx 0.002$

The two controls are mirror images. In Test A the observable is *perfectly* determined by a congruence: each class $p \bmod \ell$ maps to a single fixed value of a_ℓ, so $\eta^2 = 1.000$ exactly. In Test B the observable is *immune* to every small congruence: a_p does not track $p \bmod 12$ ($\eta^2 \approx 0.002$), the signature of Sato–Tate pseudo-randomness. Yet in both, the 6N wing ($p \bmod 3$) explains nothing.

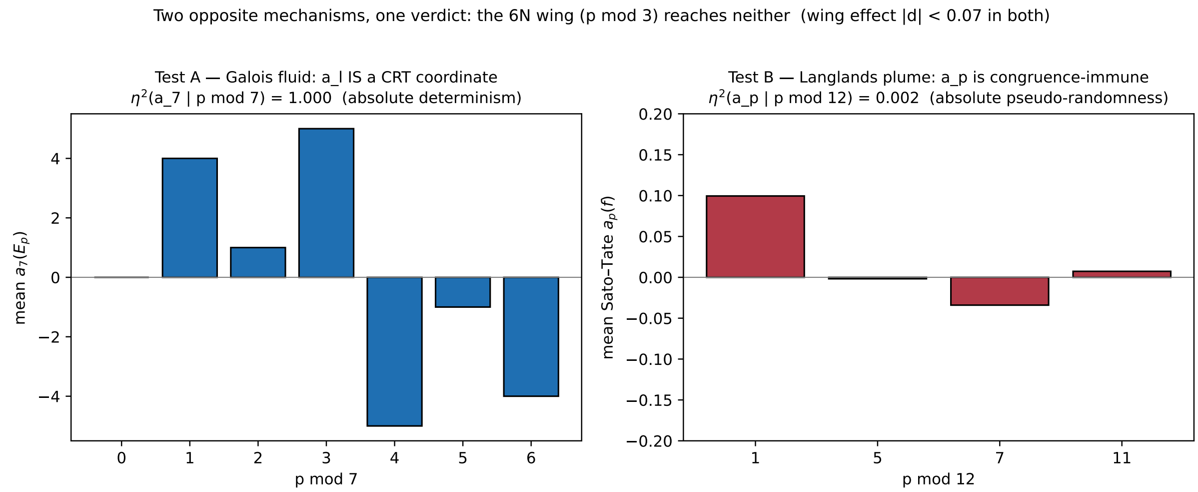

Figure 1: Two opposite mechanisms, one verdict. *Left (Test A):* $a_7(E_p)$ is a step function of $p \bmod 7$—absolute determinism, $\eta^2 = 1.000$. *Right (Test B):* the level-6 Hecke eigenvalue a_p shows no structure across $p \bmod 12$ (note the ± 0.2 scale)—absolute pseudo-randomness, $\eta^2 \approx 0.002$. In neither case does the 6N wing reach the observable ($|d| < 0.07$).

Why both fail, and what it confirms

Test A: the fluid is a coordinate. $a_\ell(E_p)$ depends only on $-p \bmod \ell$, because the reduction $E_p \bmod \ell$ does. So a_ℓ is not a deep fluid sealed inside the CRT lattice—it *is* a CRT coordinate. Bombarding p at temperature ℓ yields exactly $p \bmod \ell$, an axis orthogonal to the wing $p \bmod 3$ by the Chinese Remainder Theorem. The probe extracts the purest possible CRT product.

Test B: the gene sits on the wrong primes. The level $6 = 2 \cdot 3$ governs the *bad* primes $p = 2, 3$; it fixes a_2, a_3. For every good prime p (every $6N \pm 1$ prime) the eigenvalue a_p is a Frobenius trace whose value is congruence-immune—and in particular carries no $p \bmod 3$ information. The "6 gene" lives on the level, not on the relation between a_p and the wing. An object built on 6 still cannot make its good-prime spectrum feel which wing p lies on.

One verdict. Orthogonality of parallel coordinates (Test A) and the scale-isolation of automorphic spectra (Test B) are different physical laws, but they enforce the same boundary: the control of the 6N wing is locked into elementary number theory and the single $\chi(3)$ Euler channel. Neither a sharper extraction temperature nor a more sympathetic ambient geometry lets $p \bmod 3$ act on a good-prime L-coefficient. The $\chi(3)$-channel principle survives both attacks; having chosen the attacks to be as favourable to the wing as we could make them, we regard the boundary as confirmed.

Archival note. For Test A: `Galois_Fluid_Isotopes_Log.csv` (2260 primes), `Galois_fluid_log.sage`, `Galois_fluid_analysis.py`. For Test B: `Langlands_Plume_Log.csv` (1334 rows; $k = 4, 6$), `Langlands_plume_log.sage`, `Langlands_plume_analysis.py`. All archived at `https://github.com/Ruqing1963/6N-chi3-channel`. Each positive/mechanism control certifies its instrument ($\eta^2 = 1.000$ in A; the explicit $p \bmod 12$ table in B), so both nulls carry the highest confidence: the instruments are provably working and the wing is provably absent.